

Name: _____

TOPIC TEST

Probability

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

- 1 $A = \{2, 3, 5, 7, 11\}$, $B = \{1, 3, 5, 7, 9\}$.
Find $A \cap B$.
- A** $\{1, 2, 11\}$ **B** $\{3, 5, 7\}$
C $\{3, 5, 7, 9\}$ **D** $\{1, 2, 3, 4, 5, 7, 9, 11\}$
- 2 Which event has the same probability as tossing a tail on a coin?
- A** Choosing a vowel out of letters a, b, c, d, e, f
B Rolling 1 on a die
C Choosing a red marble from a bag of 10 red and 20 black marbles
D A family's third child being a boy

- 3 This table displays the number of cars owned by each home in Grove Ave.

Number of cars	Number of homes
0	1
1	5
2	9
3	5
4	3

Find the relative frequency that a home has 3 cars.

- A** 5 **B** $\frac{3}{23}$
C $\frac{5}{23}$ **D** 23

- 4 Elizabeth buys 5 tickets and Tania buys 6 tickets in a raffle where a total of 300 tickets were sold. Only one prize is to be won. Which expression will calculate the probability that neither Elizabeth nor Tania win the raffle prize?
- A $\frac{5}{300} \times \frac{6}{300}$
- B $1 - \left(\frac{5}{300} \times \frac{6}{300} \right)$
- C $\frac{5}{300} + \frac{6}{300}$
- D $1 - \left(\frac{5}{300} + \frac{6}{300} \right)$
- 5 The probability of a train on a city line running late is estimated at 12%. If there are 225 trains running on a city line each day, how many would be expected to arrive on time?
- A 198 B 213
- C 27 D 25
- 6 Which is the correct formula for $P(A|B)$?
- A $\frac{P(A \cup B)}{P(A)}$ B $\frac{P(A \cup B)}{P(B)}$
- C $\frac{P(A \cap B)}{P(A)}$ D $\frac{P(A \cap B)}{P(B)}$
- 7 From a group of 30 Year 11 students, 22 students study Business Studies, 9 students study Japanese (J) and 4 study neither. What is the probability that one student chosen at random from this group studies Business Studies but not Japanese?
- A $\frac{11}{30}$ B $\frac{13}{30}$
- C $\frac{17}{30}$ D $\frac{22}{30}$
- 8 A horse has a probability of 48% of winning its first race and a 36% chance of winning both its first and second race. What is the probability of the horse winning the second race, given that it wins the first race?
- A 75% B 42%
- C 17% D 12%
- 9 A box of chocolates has an equal number of dark and white chocolates. Two chocolates are chosen at random. What is the chance that at least one is a white chocolate?
- A $\frac{1}{4}$ B $\frac{1}{2}$
- C $\frac{3}{4}$ D $\frac{1}{3}$
- 10 For a double-scoop ice cream cone, Michelle will choose 2 different flavours at random from chocolate, vanilla, strawberry and mango. What is the chance that Michelle chooses chocolate and mango?
- A $\frac{1}{12}$ B $\frac{1}{6}$
- C $\frac{1}{3}$ D $\frac{1}{2}$

Part B

10 free-response questions

40 marks

Show your working where appropriate.

- 11** The 11 letters of the word MATHEMATICS are written on separate cards and placed in a bag. One card is chosen at random. What is the probability of choosing a card with the letter

a M?

b M or A or T?

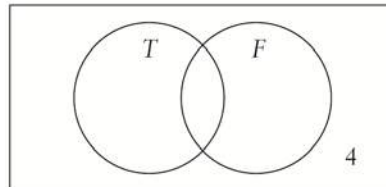
[2 marks]

- 12** A pair of dice are rolled. What is the chance that one die shows the number 2 and the other shows a number greater than 3?

[2 marks]

- 13** In a survey of 29 students, 6 are in training for both athletics track (T) and field (F) events while 14 altogether do track only and 4 students do not do any track or field events.

a Complete this Venn diagram.



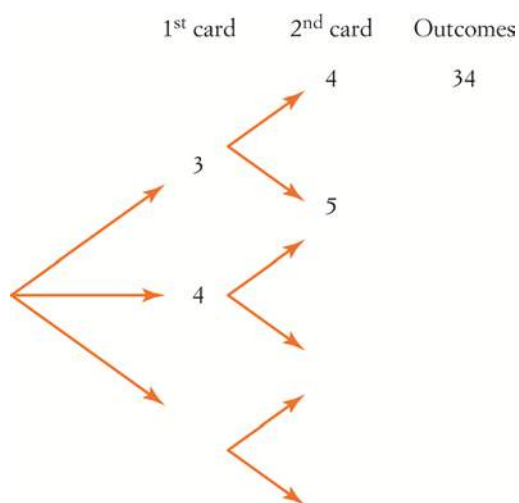
b What is the probability that a randomly-selected student does track events?

c What is the probability that, if 2 students are randomly selected, neither do any athletic events?

[4 marks]

- 14** The numbers 3, 4 and 5 are written on separate cards and placed in a bag. Two cards are drawn at random, one after the other without replacement, to form a two-digit number.

a Complete this tree diagram.



b What is the probability of forming a 2-digit number that is:

i an odd number?

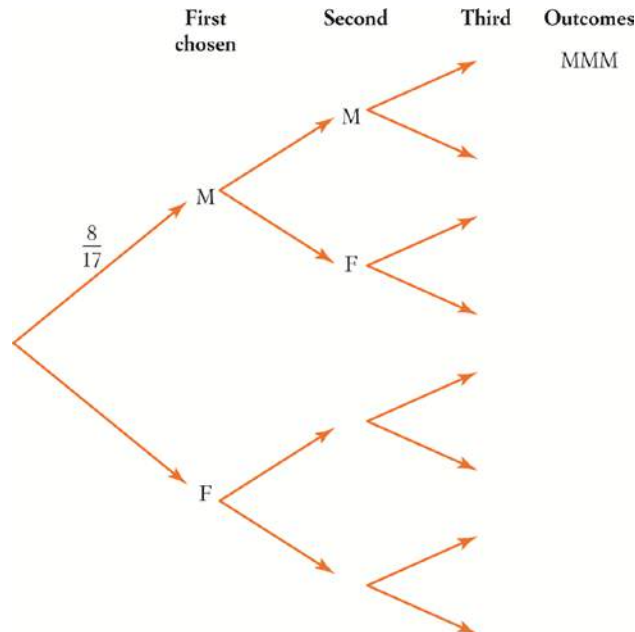
ii greater than 40?

iii a multiple of 5?

[5 marks]

15 In a class there are 8 male students and 9 female students. Three students are chosen at random.

- a** Complete the tree diagram, showing the respective probabilities on its branches and list all the possible outcomes.



- b** What is the probability that for the 3 students:

- i** all are females?

- ii** at least one is male?

[6 marks]

16 In a teacher's pencil case there are 3 black, 5 red and 2 green pens.

- a** If a pen is chosen at random, what is the probability that it is a black pen?

- b** If 2 pens are chosen at random, what is the probability that both pens chosen are red if:

- i** the first pen is replaced before the second is taken?

- ii** both pens are taken out together?

[5 marks]

- 17** Roger has a 70% chance that he will win his tennis match, while Serena has a $\frac{4}{5}$ chance of winning her match.

What is the probability that:

- a** Roger wins, but Serena doesn't win?

- b** at least one of them wins their match?

[4 marks]

- 18** A golf bag contains 6 white, 2 yellow and 4 orange golf balls. Four balls are taken without replacement. What is the probability of drawing:

- a** all orange balls?

- b** 2 white and 2 orange balls?

[4 marks]

- 19** Students at Westvale High School chose their summer sport.

Sport	Male	Female
Swimming	82	110
Cricket	61	13
Basketball	119	91
Squash	44	14
Touch football	58	65
Dance	4	33

- a** What is the ratio of male to female students who participated in this survey?

- b** What is the relative frequency of a student chosen at random playing touch football?

- c** What is the relative frequency of a student chosen at random playing touch football or basketball?

- d** What is the relative frequency of a male student chosen at random playing basketball?

[4 marks]

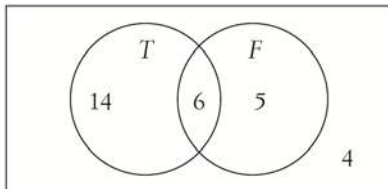
- 20** A football team has a probability of 70% of winning when Ronaldo is playing in the team. When Ronaldo is not playing, the team has a probability of 40% of winning. The probability of the team having Ronaldo playing at any one time is 90%.

Find, as a percentage correct to 2 decimal places, the probability that Ronaldo was playing when the football team won a match.

[4 marks]

This is the end of the test.

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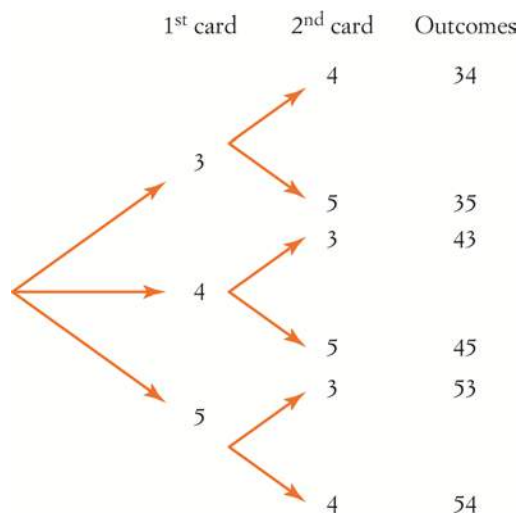
Answers**1** B**2** D**3** C**4** D**5** A**6** D**7** C**8** A**9** C**10** B**11** a $\frac{2}{11}$ b $\frac{6}{11}$ **12** $\frac{1}{6}$ **13** a

b

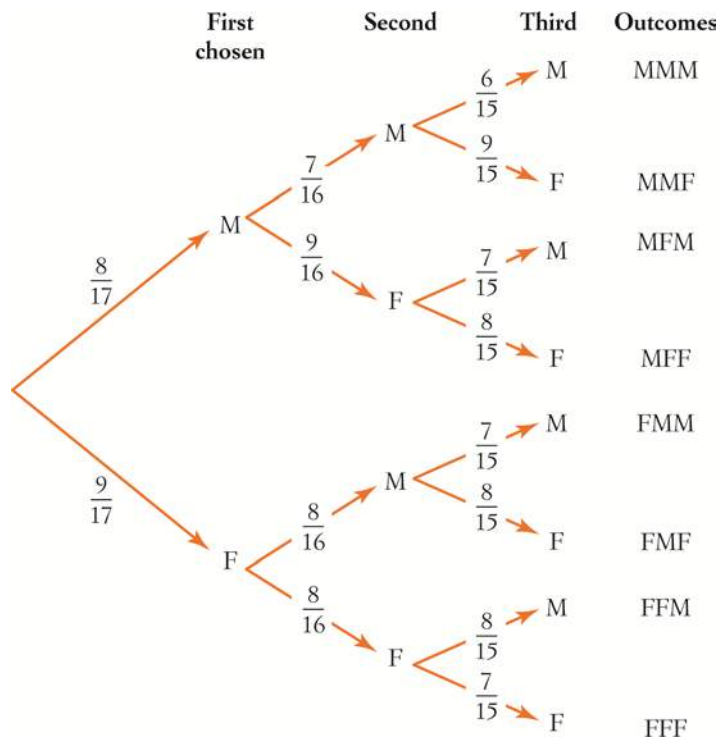
c

$$\frac{20}{29}$$

$$\frac{3}{203}$$

14 ab i $\frac{2}{3}$ ii $\frac{2}{3}$ iii $\frac{1}{3}$

15 a



b $\frac{21}{170}$

c $\frac{149}{170}$

16 a $\frac{3}{10}$

b i $\frac{1}{4}$

ii $\frac{2}{9}$

17 a 14%

b 94%

18 a $\frac{1}{495}$

b $\frac{2}{11}$

19 a 368 : 326

b $\frac{123}{694}$

c $\frac{333}{694}$

d $\frac{119}{368}$

20 $\frac{63}{67} \approx 94.03\%$